

## NONLINEAR ANALYSIS OF CENTRE SHIFT, COMPLIANCE, AND STIFFNESS OF FLEXURE PIVOTS FOR SPACE APPLICATION

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### ABSTRACT

*The use of compliant mechanisms in space applications has been discussed in literature a lot and is believed by many to have revolutionary potential in the industry, however only a couple of real-life examples could be found from the literature review. Compliant mechanisms offer benefits that would be very useful to space applications, therefore their feasibility for space needs to be analysed in detail. One type of compliant mechanisms is a flexure pivot which acts as a hinge. This report focuses on two aspects in flexure pivots: shift in centre of rotation (CoR), centre shift, and their compliance. The first flexure pivot model is a conventional Cross-Spring Pivot, while the second model is a 'Butterfly' flexure pivot. Finite Element Analysis (FEA) was done using "Abaqus FEA" to simulate some load case scenarios (LCS) that model potential real life space applications, and with the help of a "MATLAB" code the centre shift was calculated. While the compliance was analysed using eigenscrews to analyse both the kinetics and kinematics of both models. Both aspects when analysed, provided information on the accuracy, precision, and reliability of the models for space application.*

**Keywords:** Compliant mechanisms, Flexure pivots, Centre shift, Nonlinear analysis, Finite element analysis, Eigenscrews

## 1 INTRODUCTION

### 1.1 Compliant Mechanisms

Compliant mechanisms are distinctive mechanisms that are defined as movable mechanical assemblies that achieve their desired motion, force, or displacement by means of deflection of flexible members [2]. They are distinctive due to the advantages and solutions they offer to current rigid link mechanisms challenges. Such advantages include but are not limited to [5]:

1. Cheaper cost to manufacture.
2. Significant volume reduction.
3. Elimination of joints, which reduces complexity and single point failure modes.
4. Have no backlash due to their monolithic nature.
5. Ability to accurately model and predict mechanism stiffness.
6. Have increased predictability of behaviour by the use of analysis methods.

This distinction shows how compliant mechanisms are of great use as space mechanisms [12]. Especially that they do not require lubrication, have a significant reduction in moving parts which allows for less mass and a reduced risk of outgassing. Moreover, a more specific type of compliant mechanisms had to be explored for the specified space application. Firstly, Space mechanisms were categorised into single-shot deployment mechanisms and continuous mechanisms to help understand which type could be replaced by a compliant mechanism and be further analysed. Some papers were found on single-shot deployment mechanisms, and how compliant mechanisms could be used to cut cables to allow antennae to be deployed [21]. While other

papers describe continuous mechanisms such as flexure pivots in space instruments, that are mainly used for pointing and scanning [3]. Therefore, flexure pivots were a more obvious choice to explore further since they already had a suggested type of space applications to be analysed on as explained by [3].

## 1.2 Model Choice

Flexure pivots (or flexure hinges) are a type of compliant mechanisms that are used in flexure-based mechanisms since their members are important in allowing for the desired motion of said mechanism [11]. There are many different designs and models of flexure pivots that are found in literature [10, 11, 13, 19], and the choice of what models to be analysed is dependent on which models suffer the least from limitations [2] of compliant mechanisms. The limitations below are also a driving motive for the necessity to analyse the feasibility of flexure pivots in space applications.

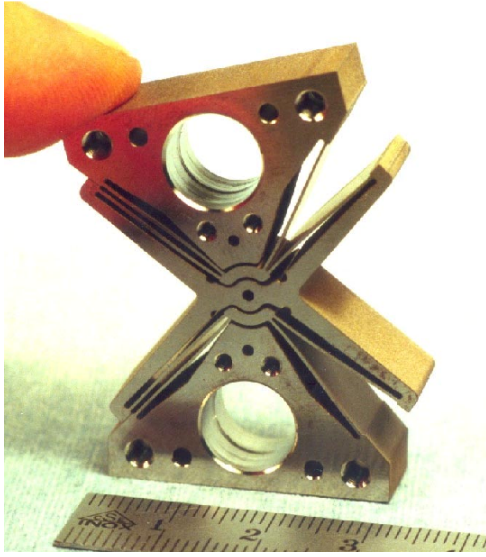
1. Simultaneous design for motion and force behaviour is required, which is hard to manipulate and design when the structure deflections are non-linear, since that is mostly the case in compliant mechanisms.
2. Fatigue life needs to surpass expected life of the mechanism due to their motion depending on bending of flexible parts.
3. Motion is more limited than that of the conventional rigid link mechanisms.
4. Storage of strain energy in deflected beams can be a disadvantage if certain materials are held under stress for long periods of time and/or elevated temperatures.

There were three main options upon consideration of the limitations and what the flexure pivot will be analysed against; the Butterfly flexure pivot [3] (Figure a), Cross-spring pivot (CSP) [4] (Figure b), and a 3D printed titanium compliant mechanism [13] (Figure e).

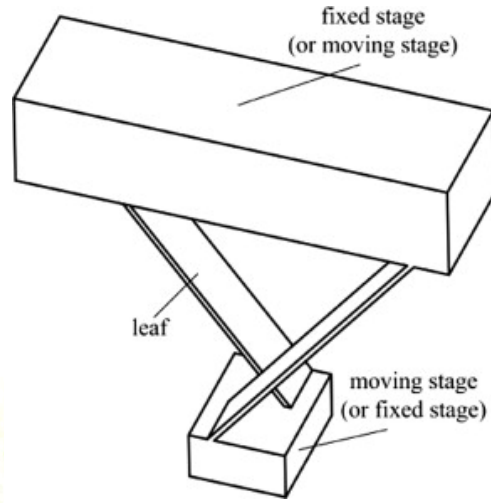
The CSP, is chosen as the first option as it has the simplest design and least to be affected by the limitations. The Butterfly flexure pivot was chosen as the second option over the titanium mechanism, due to the 4<sup>th</sup> restriction, not to mention that Henein et al, 2003, showed that the model has a very impressive fatigue life. The two models are then subjected to simulations of different load case scenarios (LCS), and nonlinear finite element analysis (FEA) was used to analyse them using "Abaqus FEA". Furthermore, a detailed MATLAB code then extracts data from the Abaqus simulation and provides both, calculation results as well as a graph of the centre shift, and visualisation of the models in terms of eigenscrews to understand their compliance. This allows for the comparison of details and the discussion of the models' feasibility on whether or not they can be used in space application to be concluded.

## 1.3 Centre Shift

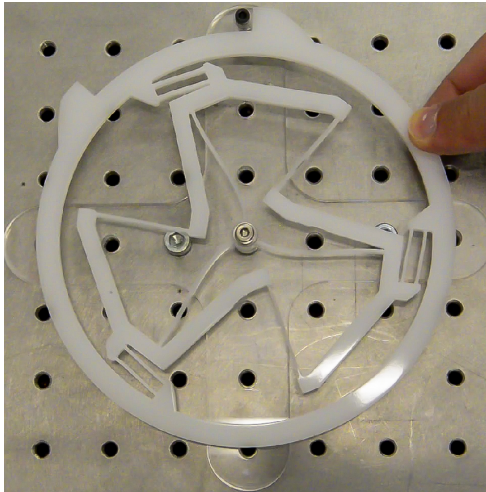
Flexure pivots are mechanisms that provide very precise rotational motion through deflection of their flexure parts [10]. This accurate rotational movement has a Centre of rotation (CoR) that can be used to find valuable information about the model, as shown in references [8, 10, 11, 18–20]. It is worth noting that some papers such as [10] and [18] refer to centre shift as "axis drift" or "parasitic shift" (short for parasitic centre shift) respectively. One of the most important ways the CoR can be exploited in, is the study of its shifting from its original position upon deflection of the structure. This is commonly known as centre shift. The centre shift is of great significance as it deteriorates the rotational guidance accuracy of flexure pivots [18], therefore the smaller the centre shift the greater the accuracy and precision of the flexure pivot.



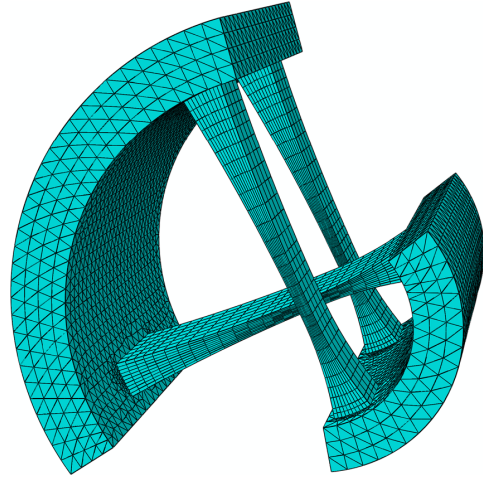
(a) Picture of the butterfly pivot courtesy of [3]



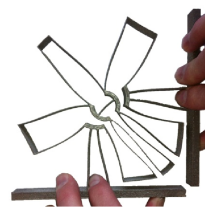
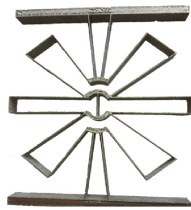
(b) Diagram of the CSP courtesy of [4]



(c) TRIVOT flexure pivot courtesy of [18]



(d) Variable Thickness Flexure Pivot (VTFP) courtesy of [19]



(e) Picture of the Titanium compliant mechanism courtesy of [13]

Figure 1: Different flexure pivot designs and models

## 1.4 Eigenscrews

Chasles' theorem states the following "the most general rigid body displacement can be produced by a translation along a line followed by a rotation about an axis parallel to that line" [1]. The lines mentioned in the theorem are often referred to as 'Screw axis', due to the displacement described being reminiscent to that of a screw [6]. Therefore, eigenscrews help to characterise

compliant mechanisms in a more physically intuitive way, as the compliant axes can be visualised using an eigenscrew decomposition. [9, 15]. Moreover, a more distinguished way to characterise compliant mechanisms is to consider the relative compliance of the principal translational and rotational axes to be characterised using eigenwrenches and eigentwists, respectively, during the deformation of the mechanism [7, 14]. Therefore, the kinetics of a mechanism can be expressed as screws of forces and moment, wrench, and the kinematics of a mechanism can be expressed as screws of combined translation and rotation, twists [17]. Thus, for a single point to be analysed such as the CoR, a 6x6 tangential stiffness matrix ( $K_t$ ) and compliance matrix ( $C_t$ ) can be formulated (1) and (2), where  $k_f$  and  $k_y$  are translational and rotational stiffness, respectively. While  $a_f$  and  $a_y$  are compliances in translational and rotational motion, respectively, and are also the inverses of  $k_f$  and  $k_y$ . Furthermore, this decomposition will result in the production of 3 eigenwrenches,  $\hat{w}_f$ , and three eigentwists,  $\hat{T}_y$ , that will describe three principal translation and rotation axes, respectively. Therefore, allowing the behaviour of a compliant mechanism to be characterised by a system of eigenscrews, a method introduced by references [9, 15]. Note: that the decomposition of the matrices below is explained in further details by Stacey et al, 2021 [17].

$$K_t = [\hat{w}_f \quad \hat{w}_y] \begin{bmatrix} k_f & 0 \\ 0 & K_y \end{bmatrix} [\hat{w}_f^T \quad \hat{w}_y^T] \quad (1)$$

$$C_t = [\hat{T}_f \quad \hat{T}_y] \begin{bmatrix} a_f & 0 \\ 0 & a_y \end{bmatrix} [\hat{T}_f^T \quad \hat{T}_y^T] \quad (2)$$

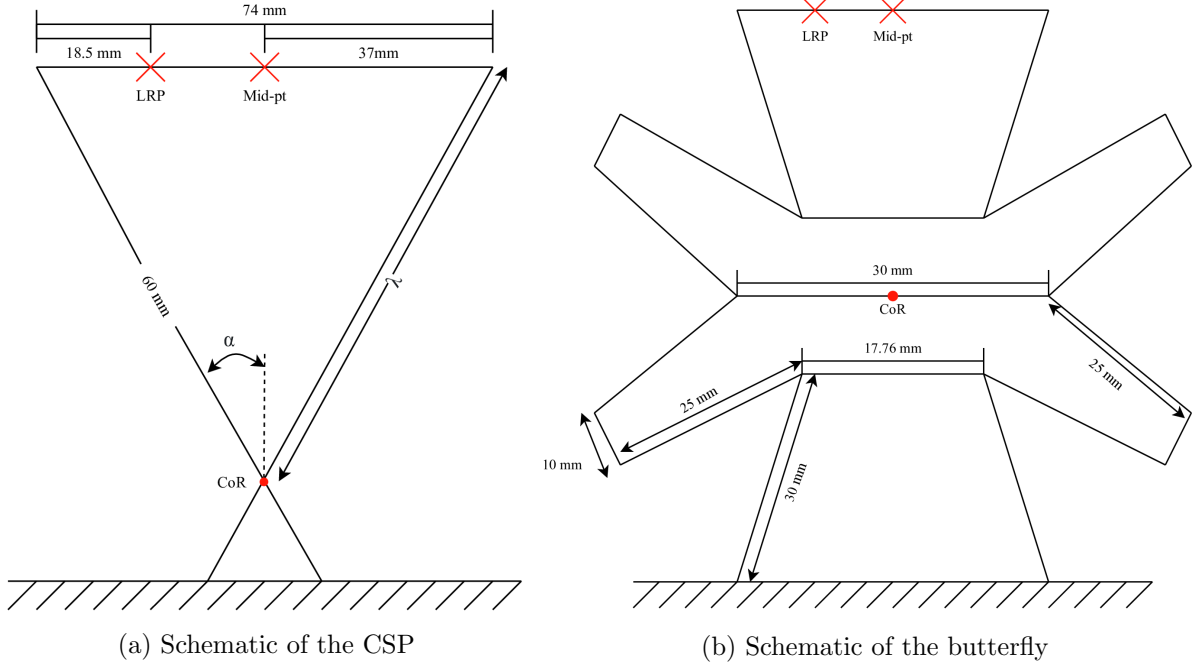


Figure 2: Schematics of both flexure pivots

## 2 METHODOLOGY

### 2.1 Experimental Limitations

The position of the CoR differs with each model, since it is the centre of rotation of the Mid-pt in the structure that is being analysed. For the Butterfly model the CoR of the Mid-pt is located in the centre of the model. While, for the CSP it is located at the intersection point of both

leaves (note that the CoR location need not be on the physical structure of the model [18]). As mentioned above centre shift occurs during the deflection of the structure, and the compliance needs to be analysed for deformation of the structure. However, for this research it was not feasible financially nor time efficient to manufacture the models with the required material (Aluminium). Moreover, it would not have been scientifically accurate to 3D print the models using PMMA (Poly(methyl methacrylate)) Plastic, as that would prevent comparison of results with sources that built the models, such as [3] and the butterfly pivot.

## 2.2 Finite Element Analysis (FEA)

### 2.2.1 Software Choice

The next best option was to simulate the different load case scenarios to the models on a FEA software. Since the analysis of the centre shift are nonlinear, "Abaqus FEA" was the software of choice as it is renown for its nonlinear analysis and user-friendly interface. The latter advantage was of importance as the author had to learn and get accustomed to the software in a short time span. Note, the software was used under the license acquired by the University of Bristol.

### 2.2.2 Designing the Simulations

#### Flexure Pivot Models

Upon testing and modeling of many preliminary models on Abaqus it was deduced that building both flexure pivots as beams connected end-to-end would be the best and most accurate choice. A MATLAB code was used to generate a .INP file (Abaqus input file) to be used as a source for jobs in the software. Both models were subjected to an initial boundary condition of *Encastre* on the tips of two beams (as shown in (Figure 3) this boundary condition prevents any translation or rotation on those ends ( $U1=U2=U3=UR1=UR2=UR3=0$ ), hence creating a fixed stage for the pivot. Beam parts were connected using *Connector Sections* which allows for all 6 degrees of freedom (DOF) to be constrained with the other beam. All beams were assigned the material of Aluminium with an Elastic Modulus = 69 GPa, Poisson Ratio = 0.33, and a Density of  $0.0027 \text{ tonne/mm}^3$  ( $2700 \text{ kg/m}^3$ ). Schematics showing dimensions of both flexure pivots can be seen in Figure 2, the width of the CSP beams is 6 mm and the width of the butterfly beams is 5 mm. When it comes to thickness however, thicker beams and payloads in both models were made stiffer by making them 6x thicker than the other beams. The thick beams and payloads were modeled with 6mm thickness while thin beams were modeled with 0.5 mm thickness. Furthermore, in the CSP model  $\lambda$  stands for the ratio of the leaves before the intersection point, in this case  $\lambda = \frac{1}{2} + \sqrt{\frac{5}{6}}$ , and  $2\alpha$  stands for the angle of intersection, in this case  $\alpha = \frac{\pi}{4}$ . Dimensions of the CSP are specified as those of the simulations done by [4] to allow for comparison in results and data obtained. Furthermore, the butterfly pivot was simplified from the structure seen in a to that seen in the schematic b. This simplification was done with guidance from the project supervisor and with help from literature, as it was done before in [16]. This idea behind the simplification is to only include vital beams for the rotational motion in the design, and making some beams stiffer than others by making them thicker (6x thicker in this case). Finally, a global meshing size of 0.5 is used for both designs, therefore the butterfly pivot has a total of 1001 nodes and 980 elements, while the CSP has a total of 392 nodes and 388 elements.

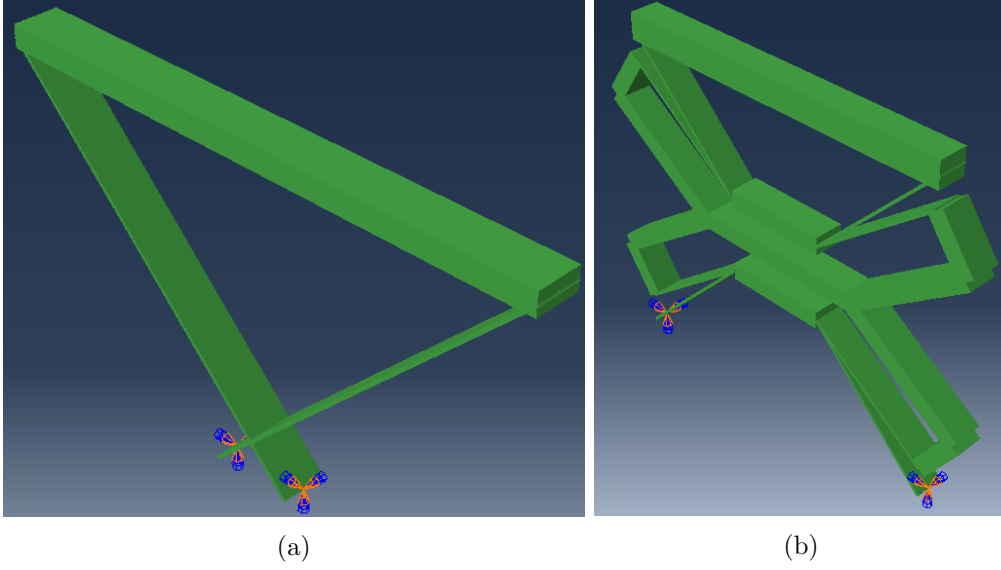


Figure 3: *Encastre* boundary condition on the CSP (a) and on the butterfly pivot (b)

### Load Case Scenarios (LCS)

The simulation is used to model a real life application on the pivots. This means if a force was to be applied on the pivot the simulation was used to model it to see its effects on the structure of the pivot. As mentioned before, flexure pivots are used for precise, accurate rotational motion. Therefore, the real life applications to be considered were real life pointing mechanisms in space. After literature review, the author believes that flexure pivots could be used for solar panels pointing mechanisms, as those do not require big deflection angles, and the flexure pivots are modelled to provide  $\pm 15$  deg according to literature [4,16]. Not to mention, that the weight of the solar panels will not be a factor to consider on the structural loads applied in space. Moreover, said literature results will be used for comparison with the results produced in the report, as a method of result verification. It is worth noting, that the actual detailed application is of no interest here, and that only a generalised concept needs to be considered for simulation purposes to study the feasibility of the pivots. To model the pointing mechanism, an axial force was applied in the 1<sup>st</sup> direction (X-direction, U1) and a moment was applied about the 3<sup>rd</sup> direction (Z-direction, UR3). Both a moment and a force are used because when coupled together, they are similar to a force applied at the middle of the payload (4). The forces were applied at two main points during the simulations, Middle Point (Mid-Pt) and Left Reference Point (LRP) to test the structural integrity of the pivots and also to make the simulations as realistic as possible since not all forces will be applied at the middle in real life application. For both the Butterfly and the CSP Mid-pt is located in the middle of the beam modelling the payload, while the LRP is also in the same position in both pivots. It is equidistant from both the Mid-pt and the left edge of the beam modelling the payload. The 6 different LCS that were simulated using FEA can be seen in Table 1 (note: CF1 and CM3 stand for Concentrated Force in 1<sup>st</sup> direction and Concentrated Moment in 3<sup>rd</sup> direction respectively). The loads that were chosen in the LCS are chosen after many analyses were run to test the limits of the pivots, but it was assumed that the initial load was applied at a vertical distance (L) of 10 mm. All forces were modelled to follow nodal direction to ensure they stay perpendicular to the to CoR across the whole time they are applied. Both the forces and the moments were simulated in the same time step in the FEA, the time step is set as a static, general step with a maximum number of 50 increments. Increments were set to have a maximum size of 0.02 and a minimum size of  $1 \times 10^{-5}$ , this allowed Abaqus to calculate the number of increments used in a simulation.

| Load Case Scenarios |           |               |       |                    |        |           |                 |                    |
|---------------------|-----------|---------------|-------|--------------------|--------|-----------|-----------------|--------------------|
| Set                 | Direction | Magnitude (N) | Type  | Application Region | Type   | Direction | Magnitude (Nmm) | Application Region |
| 1 <sup>st</sup>     | CF1       | 1             | Force | LRP                | Moment | CM3       | -10             | Mid-Pt             |
| 2 <sup>nd</sup>     | CF1       | 1             | Force | Mid-Pt             | Moment | CM3       | -10             | Mid-Pt             |
| 3 <sup>rd</sup>     | CF1       | 5             | Force | LRP                | Moment | CM3       | -50             | Mid-Pt             |
| 4 <sup>th</sup>     | CF1       | 5             | Force | Mid-Pt             | Moment | CM3       | -50             | Mid-Pt             |
| 5 <sup>th</sup>     | CF1       | 7.5           | Force | LRP                | Moment | CM3       | -75             | Mid-Pt             |
| 6 <sup>th</sup>     | CF1       | 7.5           | Force | Mid-Pt             | Moment | CM3       | -75             | Mid-Pt             |

Table 1: The 6 LCS details, note that the moment magnitude is negative to match the direction of the force.

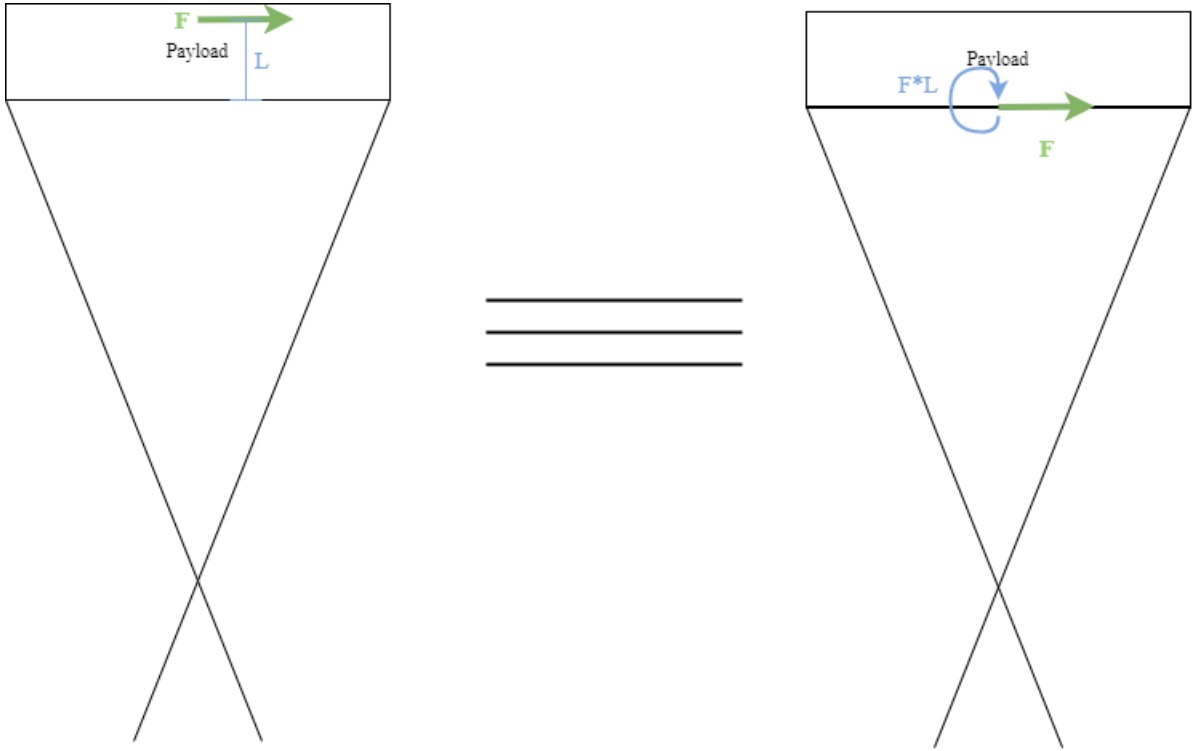


Figure 4: Schematic showing the relationship between coupling of a force and a moment to a force applied in the middle of the payload.

## Data Extraction

FEA data of the Mid-Pt is extracted by commanding Abaqus to generate a history output for certain variables of the node. Such variables were: U1, U2, UR3, CF, RM3 (reaction moment in 3<sup>rd</sup> direction) and COORD (coordinates of the point) (COORD was extracted for not only the Mid-pt but all nodes). Moreover, a stiffness matrix was generated, but this will be discussed thoroughly in another section. The output variables were then saved in a Data (.DAT) file that was used by a MATLAB code for analysis.

## 2.3 Analysing the Centre Shift

To analyse the Centre Shift in both pivots, four main variables needed to be saved, U1, U2, UR3 and RM3. The difference between each increment and the other in every variable is then found;  $\Delta U1$   $\Delta U2$   $\Delta UR3$ . Afterwards, the relative position vectors (in the X and Y directions) of



CoR with respect to the position of the Mid-pt at the current increment ( $R_x$  and  $R_y$ ) would be calculated using equations (3) and (4). Then an important step after that will be to account for the displacement of the Mid-pt, this is important as it allows for the position vectors of the CoR with respect to the original position of the Mid-pt ( $D_{x_c}$  and  $D_{y_c}$ ), which represents the centre shift in the X and Y directions, to be found. This step was done for every single increment as well using equations (5) and (6), where U1, U2,  $R_x$ , and  $R_y$  are all values at the same increment during the calculation. However, it is very important to account for the original coordinates of the Mid-pt when calculating  $D_{x_c}$  and  $D_{y_c}$ . For example, if the original coordinates of the Mid-pt were (2,3) then equations (7) and (8) shall be used, where  $U_{1_0}$  and  $U_{2_0}$  in this case will be 2 and 3 respectively. But if the original coordinates were (0,0), this step shall be ignored.

$$R_x = \frac{U1}{UR3} \quad (3)$$

$$R_y = \frac{U2}{UR3} \quad (4)$$

$$D_{x_c} = U1 - R_x \quad (5)$$

$$D_{y_c} = U2 - R_y \quad (6)$$

$$D_{x_c} = U_{1_0} + D_{x_c} \quad (7)$$

$$D_{y_c} = U_{2_0} + D_{y_c} \quad (8)$$

Finally, the graphs of  $D_{x_c}$  and  $D_{y_c}$  against RM3 were plotted for the CSP while for the Butterfly the X-axis variable was UR3, where RM3 is the moment about the Mid-pt being deflected and UR3 is the rotational deflection of the Mid-pt about the Z axis. The reason behind the difference in the X-axis variables is simply to compare with literature, as both variables could be used to analyse the centre shift.

## 2.4 Analysing Compliance using Eigenwrenches and Eigentwists

### 2.4.1 Data Extraction

The process of analysing the compliance of the flexure pivots is very similar to that of the centre shift, while the data obtained from the analysis is very different. It was mentioned above in section 2.2.2, in the 'Data Extraction' paragraph that a stiffness matrix (.MTX file) is generated as part of the simulation results. While the stiffness matrix is beneficial, it alone was not sufficient to analyse the compliance of the pivots. Therefore, a complex and detailed MATLAB code was used to process the matrix and the data file post simulation. Firstly, the code would read and interpret the Stiffness matrix and use it to calculate the eigenwrenches;  $w_1$ ,  $w_2$  and  $w_3$  and eigentwists;  $t_1$ ,  $t_2$  and  $t_3$ . Afterwards, the code would then read the data file (.DAT) and extract the coordinates of all nodes, displacement in the X and Y directions (U1 and U2) and the load applied to the structure (CF1 and CM3). Finally, the code would then produce 3D plots visualising the deformed and undeformed structure of the flexure pivot alongside all 6 eigenwrenches and eigentwists to show the direction of each.

### 2.4.2 Understanding the Data

In the previous section, it was explained how eigenwrenches and eigentwists values will be obtained from the FEA simulation results, but what do the values actually mean? To understand what results to expect and what the values actually mean, a simple simulation was performed to



help interpret the results. The simplest idea is to attach a cantilever beam to a wall (using the Encastre boundary condition in FEA) and apply a point load,  $P$ , to the very tip of the beam, which will be called POI, short for point of interest (POI in the other models is the Mid-pt). The same methodology to extract and obtain the data was used as explained in the section above. Figure (5) shows the 3D plot of the cantilever beam with traces of its undeformed shape, and with all the eigenwrenches and eigentwists plotted and in table (3) the values of all 6 variables of this simulation can be found. The first step of understanding what the values in table (3) represent is applying some 'back of the envelope' equations; equations (9) and (10), where  $P$  is the point load,  $L$  is the length of the cantilever beam,  $E$  is the elastic modulus (or young's modulus),  $I$  is the second moment of area,  $\theta$  is the end slope and  $\delta$  is the maximum deflection of the beam (at the tip). The simulation data of the variables above is listed in table (2). The equations and the data obtained would then be compared with other data when  $P$  is varied. This would show how the eigenwrenches would differ in value along with a decrease or increase in both  $\theta$  and  $\delta$ . From the results and values shown it was deduced that for eigenwrenches a low value is desired for the principal wrench as it shows compliance, while for eigentwists a larger value will be required for the principal twist as it represents stiffness.

$$\theta = \frac{PL^2}{2EI} \quad (9)$$

$$\delta = \frac{PL^3}{3EI} \quad (10)$$

| Set | P (N/m) | L (m) | E (Pa)           | I ( $m^4$ )            | $\theta$ (rad) | $\delta$ (m) |
|-----|---------|-------|------------------|------------------------|----------------|--------------|
| 1   | 0       | 0.074 | $69 \times 10^9$ | $1.35 \times 10^{-11}$ | 0              | 0            |
| 2   | -25     | 0.074 | $69 \times 10^9$ | $1.35 \times 10^{-11}$ | -0.0736        | -0.0036      |
| 3   | -50     | 0.074 | $69 \times 10^9$ | $1.35 \times 10^{-11}$ | -0.1472        | -0.0073      |
| 4   | -100    | 0.074 | $69 \times 10^9$ | $1.35 \times 10^{-11}$ | -0.2943        | -0.0145      |

Table 2: Cantilever simulation data and results

| Set | W1     | W2      | W3       | T1       | T2       | T3       |
|-----|--------|---------|----------|----------|----------|----------|
| 1   | 0.0365 | 0.00927 | 5.96e-05 | 1.99e-05 | 7.77e-05 | 7.95e-05 |
| 2   | 0.0365 | 0.00927 | 6.31e-05 | 7.76e-05 | 1.99e-05 | 7.95e-05 |
| 3   | 0.0365 | 0.00929 | 7.33e-05 | 7.74e-05 | 2.00e-05 | 7.96e-05 |
| 4   | 0.0365 | 0.00934 | 1.14e-04 | 7.66e-05 | 2.02e-05 | 7.99e-05 |

Table 3: Eigenwrenches and eigentwists values in different simulations on the cantilever beam

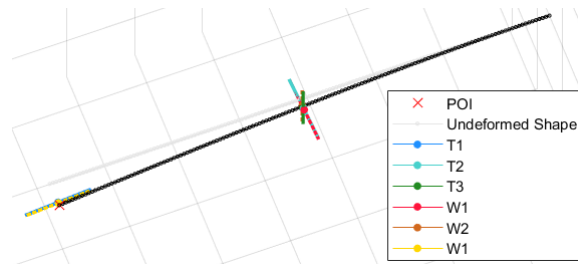


Figure 5: 3D plot of the deformed cantilever beam with all eigenwrenches and eigentwists plotted

### 3 RESULTS

#### 3.0.1 Centre Shift Results

As explained in the Methodology section, the results of the centre shift analysis are different than those of the compliance analysis. For the centre shift the results expected are plots of the shift of the CoR in both X and Y directions ( $D_{x_c}$  and  $D_{y_c}$ ), for every LCS, against either RM3 or UR3, for the CSP and the butterfly, respectively.

#### Cross-Spring Pivot

Below are all 12 plots of  $D_{x_c}$  and  $D_{y_c}$  against RM3 for the CSP model. The graph shows a

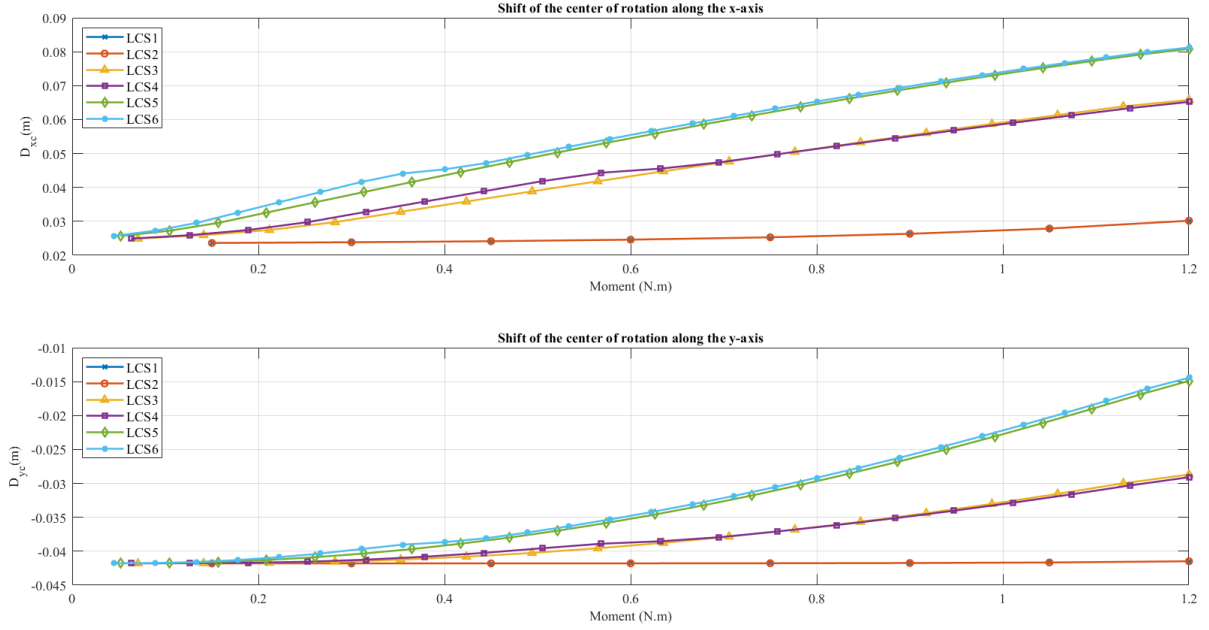


Figure 6: Plots of  $D_{x_c}$  and  $D_{y_c}$  against the reaction moment about the 3rd direction, for all 6 load case scenarios applied on the CSP.

trend of increasing and decreasing centre shift in both the X and Y direction ( $D_{x_c}$  and  $D_{y_c}$ ), respectively, as the moment increases. It is worth noting that, for  $D_{y_c}$  a greater value numerically means less deflection in the Y direction as the centre shift is negative due to its direction (in this case downwards relative to figure (8)). It is also shown that the greater the load (overall) the greater the shift in the CoR as the LCS with greater concentrated force (CF) and concentrated moment (CM) lead to the CoR to shift to more in the 1st and 2nd directions. Furthermore, the results fail to conclude a trend between when the loads are applied on the centre of the payload (on the Mid-pt) or on the LRP.

With respect to  $D_{x_c}$  figure (7) shows that for LCS 1 and 2, when the CF is applied at the Mid-pt, a slightly higher centre shift will be produced. Moreover, the results in figure (6) show that LCS 5 and 6 follow a similar trend but with a bigger, and more noticeable difference in  $D_{x_c}$ . Meanwhile, LCS 3 and 4 show an opposing trend to the other four.

However, when it comes to  $D_{y_c}$  similar trends are seen with respect to LCS 1, 2, 3 and 4 as all four show that as the CF is applied at the Mid-pt a more negative  $D_{y_c}$  is produced for the same value of Cf and CM as seen from figures (7) and figure (6). Yet LCS 5 and 6 show an opposing trend to that of the other four, as LCS 6 has a less negative centre shift than that of LCS 5. All

this contradiction leads to failure of deducing an obvious trend when it comes to positioning of the CF in the cross-spring pivot.

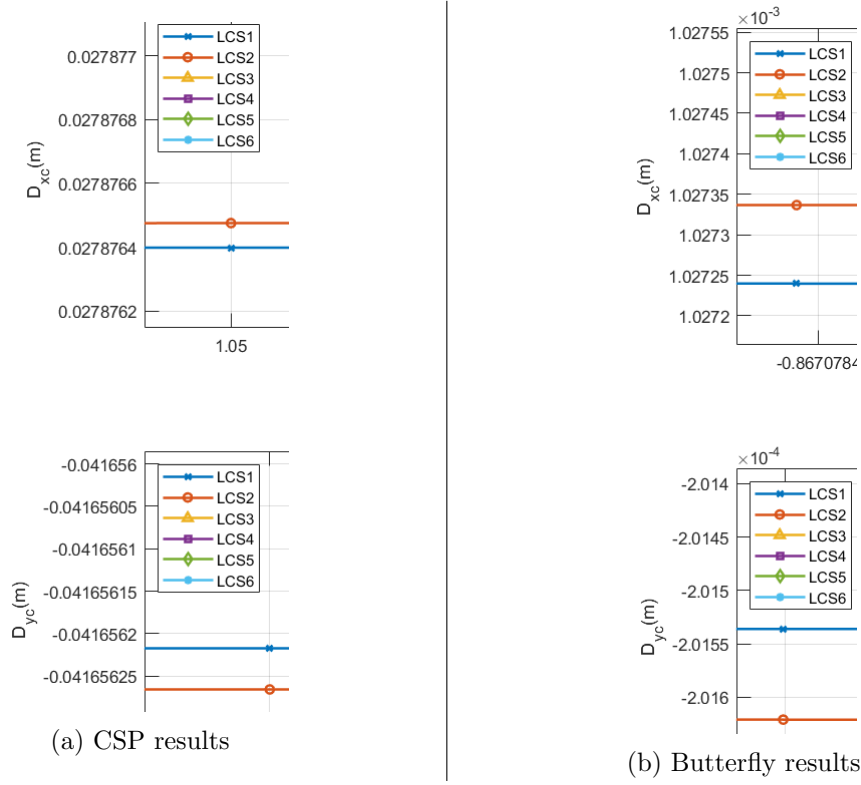


Figure 7: A closer look on the results of both flexure pivots.

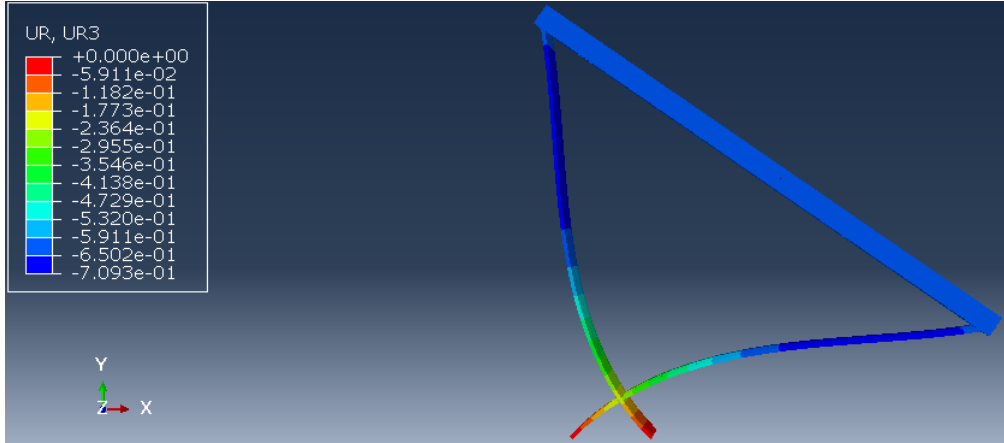


Figure 8: Visualisation of the deflection caused by LCS 4 on the CSP structure, note that UR3 here is in rad.

### Butterfly Flexure Pivot

A graph can be seen in figure (9) including all 12 plots of  $D_{xc}$  and  $D_{ye}$  against UR3 for the butterfly model. While running the simulations of LCS 3-6 on the Butterfly model, the structure would tend to overlap in an interesting and unexpected way as seen in figure (10) for LCS 6 specifically. This meant that the for LCS 3-6 the motion range was limited by the structure

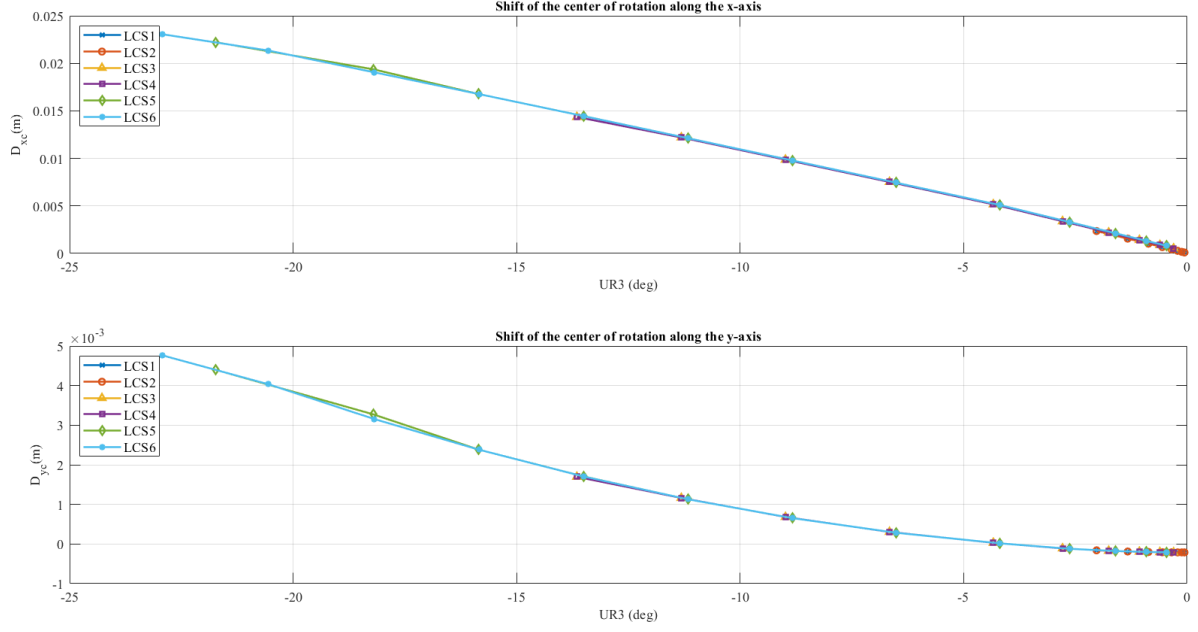


Figure 9: Plots of  $D_{xc}$  and  $D_{ye}$  against the rotational deflection of the Mid-pt about the 3<sup>rd</sup> direction (Z axis), for all 6 load case scenarios applied on the butterfly flexure pivot.

instead of the yield stress of Aluminium (276 MPa), which is normally the limiting factor for range in flexure pivots, as the structure starts to overlap before reaching yield stress. In figure (10) overlapping occurs at around 207 MPa. Therefore, for the rest of this section the results of LCS 3-6 will be ignored and will be commented on in more details in section 4.

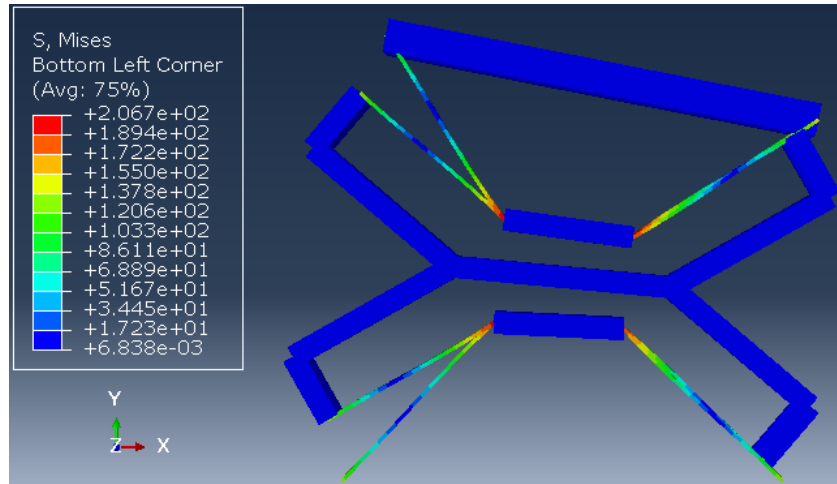


Figure 10: Visualisation of the unusual deformation undergone by the Butterfly pivot in LCS 6, note that the Von Mises stresses,  $S$ , here is in MPa.

To allow for better understanding of the results the graph was re-plotted (figure (11)) with only the results of LCS 1 and 2. Both LCS show almost identical results, concluding that for the butterfly flexure pivot, the CF position has negligible effects on the centre shift (this can be further seen in figure (7b)). In addition,  $D_{xc}$  sees an increase as the structure is further deflected, while  $D_{ye}$  has an opposite trend (shift is in the negative direction so less negative means less shift). Finally, in both flexure pivots there was no LCS that produced a perfectly linear relationship between the two plot variables.

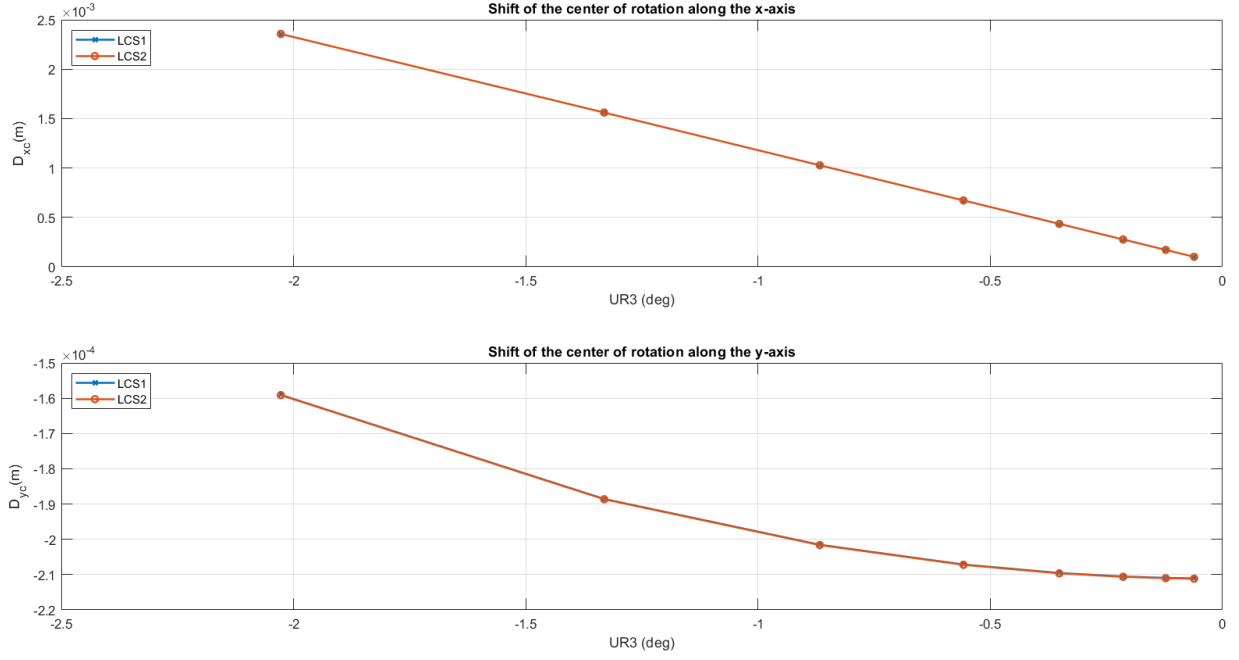


Figure 11: Plots of  $D_{xc}$  and  $D_{y_c}$  against the rotational deflection of the Mid-pt about the 3<sup>rd</sup> direction (Z axis), for LCS 1 and 2 applied on the butterfly flexure pivot.

### 3.1 Compliance Results

#### Cross-Spring Pivot

In figure (12) 6 3D plots of the CSP under each LCS can be found. The results are obtained with the help of the Abaqus solution and the MATLAB code.

Table (4) summarises all the eigenwrench and eigentwist values for all 6 LCS applied to the CSP. From LCS 1 - 4 an almost identical trend is shown where W2 and W3 have the smallest magnitudes and hence most compliance, while T3 has the biggest magnitude of all eigentwist values. Therefore the CSP is least stiff about its axis (T3 has principal axis). However for LCS 5 and 6, all three eigenwrenches tend to switch orientations (see figures (12e) and (12f)) while their magnitudes remain the same, whilst all eigentwists remain unchanged in orientation.

| Set | W1    | W2     | W3     | T1       | T2       | T3      |
|-----|-------|--------|--------|----------|----------|---------|
| 1   | 0.138 | 0.0126 | 0.0121 | 6.29e-05 | 4.62e-04 | 0.00140 |
| 2   | 0.138 | 0.0126 | 0.0121 | 4.62E-04 | 6.28e-05 | 0.00140 |
| 3   | 0.140 | 0.0209 | 0.0115 | 4.94e-04 | 6.92e-05 | 0.00160 |
| 4   | 0.140 | 0.0209 | 0.0115 | 4.93E-04 | 6.88e-05 | 0.00140 |
| 5   | 0.143 | 0.0321 | 0.0109 | 5.36e-04 | 7.76e-05 | 0.01170 |
| 6   | 0.143 | 0.0322 | 0.0109 | 5.34e-04 | 7.70e-05 | 0.01140 |

Table 4: Eigenwrenches and eigentwists values in different simulations on the CSP model

With respect to the position of the CF, a very noticeable trend can be seen in table (4) for all six LCS, where the positioning does not effect the magnitudes of the eigenwrench components.

Moreover, when positioned on the Mid-pt, the values of all 3 eigtwist tend to decrease and hence the structure tends to be more stiff about all three axes. One final thing to mention about the compliance of the CSP is that, T1 and T2 switch orientation between LCS 1 and 2 and then switch back in LCS 3. Yet, their magnitudes are switched between LCS 1 and 2 and never returned.

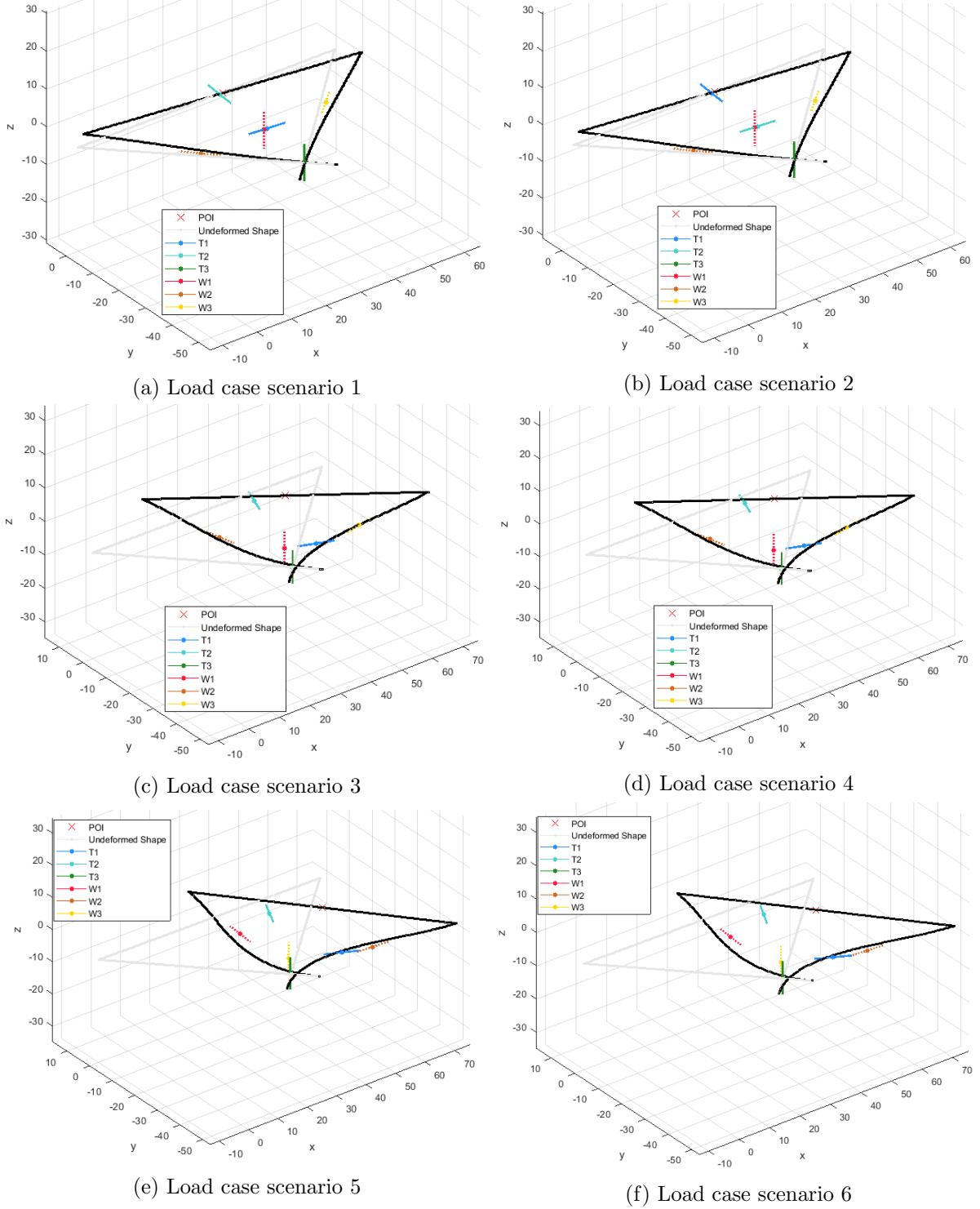


Figure 12: Visualisation of all 6 load case scenarios showing the direction of each eigenwrench and eigtwist and the deflection of the CSP structure, Note: POI here represents the Mid-pt.

### Butterfly Flexure Pivot

Below are all 6 3D plots of the Butterfly under each LCS (figure(13)), obtained with the help of the Abaqus solution and the MATLAB code. A table summarising the magnitudes of all eigenwrenches and eigentwists was not produced as coincidentally both flexure pivots produced the same exact results. Hence, table(4) will be used for both flexure pivots.

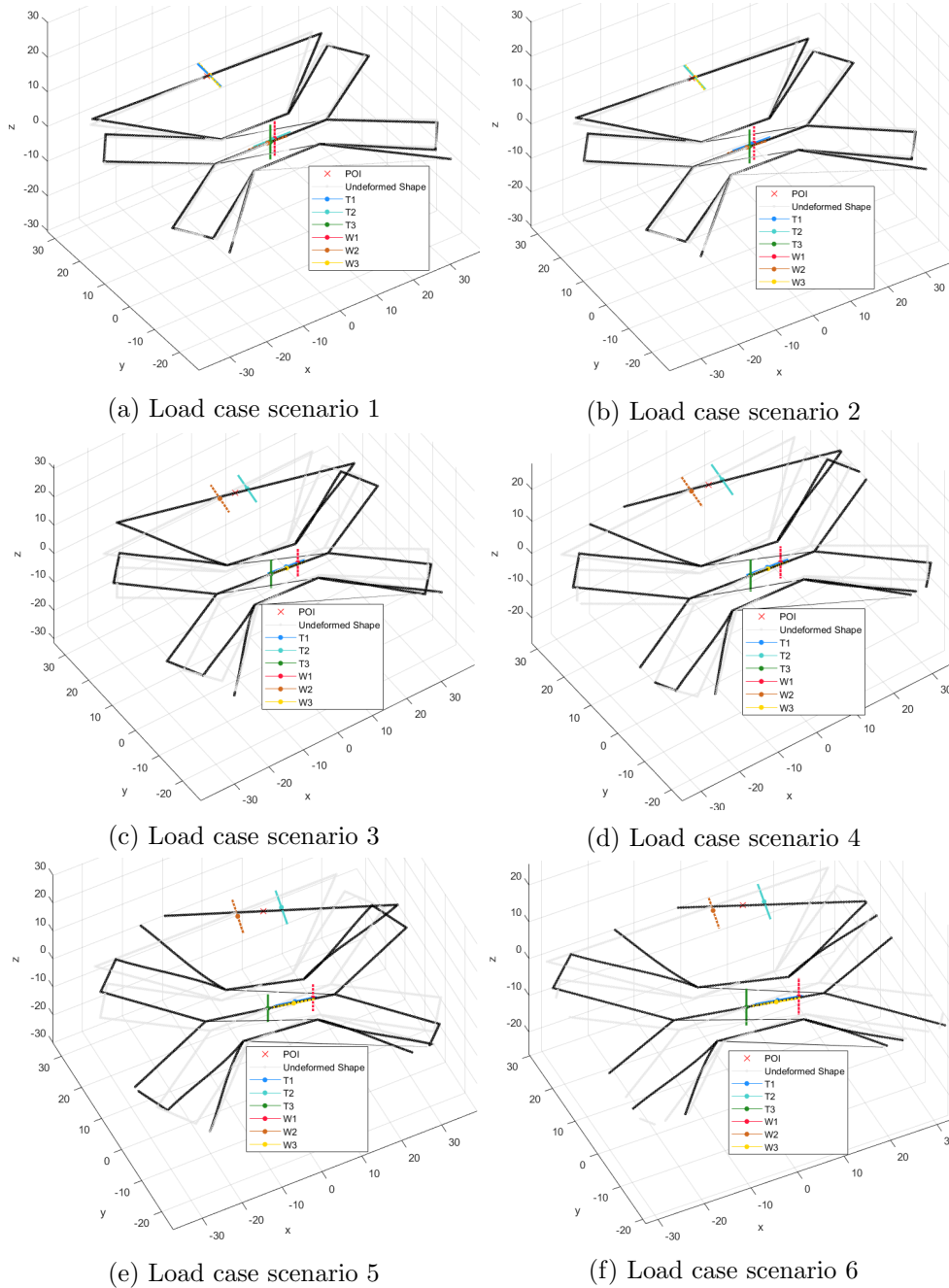


Figure 13: Visualisation of all 6 load case scenarios showing the direction of each eigenwrench and eigentwist and the deflection of the butterfly structure, Note: POI here represents the Mid-pt.



Across all 6 simulations W2 and W3 showed to be the most compliant axes for the flexure pivot, with W3 being more compliant by a small difference that increased as the overall load increased. T3 proved to be the axis with least stiffness to revolve around and its orientation remained unchanged for all simulations. Moreover, the orientation and values of T1 and T2 flipped after LCS 1 but remained unchanged onwards from LCS 2 to 6, unlike the CSP where the two axes switched orientations again at LCS 3, see figures (13a) and (13b).

## 4 DISCUSSION

### 4.0.1 Cross-Spring Pivot

The CSP results provided ample insight on the model. Initially a comparison between the obtained results and literature has to be analysed. In [4] the authors applied a vertical downward force,  $p$ , at the CoR of the CSP as modeling of the weight of the payload. However, in the case of space applications this force will not exist, and thus it is not applied in this report.

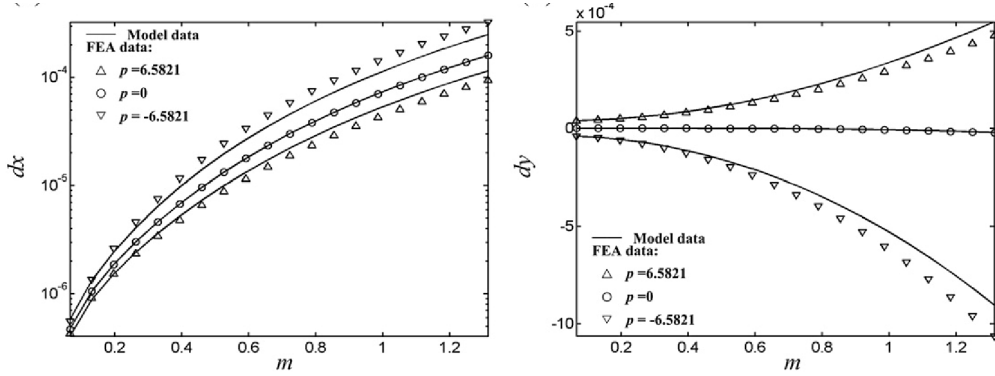


Figure 14: Data obtained from literature for the centre shift in the CSP model, courtesy of [4]

In figure (14)  $dx$  and  $dy$  are equivalent to  $D_{x_c}$  and  $D_{y_c}$  and are also in meters, while the  $m$  on the X-axis stands for moment and is also in (N.m). Therefore for the plot of 'p=0' there is an obvious difference between the results obtained and those in the literature, the most noticeable is the difference in the centre shift values. Firstly, centre shift in the 2nd direction is completely zero when there is no vertical force applied which is not the case in the results obtained, see figure(6). Additionally, the centre shift in the X direction shows a different trend in the plot and from the obtained results has much larger values. The author hypothesises that this difference in plots might be the result of different calculation methods on the extracted FEA data. Since even if it was a unit conversion error, the difference in trend remains unexplained even with the difference in data points. Not to mention, literature review did not unveil a method of calculation for the extracted FEA data, and that all the calculations shown were analytical solutions driven from geometry. Ultimately, whilst the graphs in figure (6) failed to show an obvious trend between LCS with the same overall load but different application regions, the difference between each plot of LCS was not noteworthy. Thus, suggesting that the cross-spring pivot will not be highly effected (precision wise) if in real life application the load was not aligned with the structure.

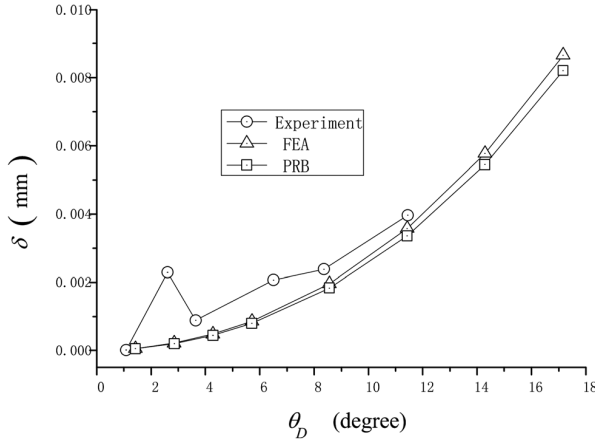
After weeks of trials, simulations, consulting the supervisor and contacting authors (Li at al, 2022 were contacted for enquiries [8]), the comparison between the literature and the obtained results had to be left as is. Moreover, the results of the CSP that were obtained still proved to be enthralling. An obvious deduction is that the limiting factor on the range of motion of the CSP is the yield stress of Aluminium and not the structure, hence the flexure pivot provides a great range of motion when the Von Mises stresses are under the yield stress. Furthermore, when it comes to compliance and stiffness analysis, the CSP showed compliance about W2 and

W3 (with W3 being the most compliant axis). Which up until LCS 4, meant that the pivot was being compliant in directions that benefits the range of motion, however afterwards, (see figures (12e) and (12f)) all the eigenwrenches changed orientations while maintaining their magnitudes. This suggests that under loads as high as those of LCS 5 and 6 (see table (1)) when the CSP is under high deformation, the structure tends to show compliance in undesirable directions (Z-direction in this case).

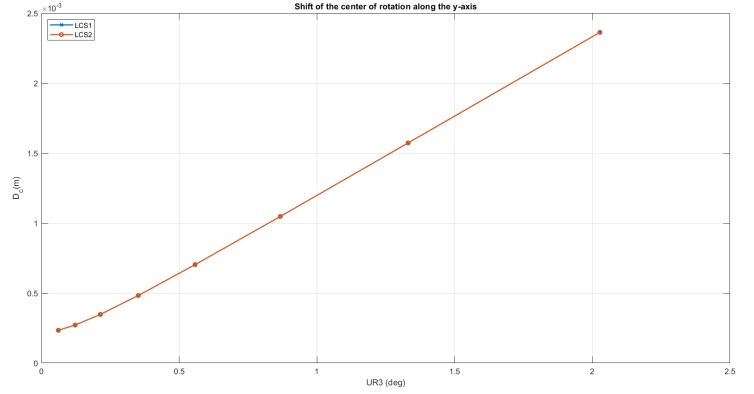
#### 4.1 Butterfly Flexure Pivot

The butterfly flexure pivot results were compared with that of a similar structure ([16], figure (6.b)) found in the literature. However, to allow for a comparison in the results,  $D_c$  was calculated using equation (11) and was plotted (figure (15b)) against a positive UR3 (results are symmetrical due to nature of the structure).

$$D_c = \sqrt{D_{x_c}^2 + D_{y_c}^2} \quad (11)$$



(a) Centre shift  $\delta$  against rotational angle  $\theta$  for the butterfly pivot, courtesy of [16]



(b)  $D_c$  against the rotational angle, UR3, for the butterfly pivot under simulations of LCS 1 and 2.

Figure 15: Comparison between obtained results and literature for the butterfly pivot centre shift.

The obvious similarity in trend confirms that the method of CoR shift (centre shift) calculation is correct. Nonetheless, there is a noticeable difference in magnitude of the values of

$$D_c$$

around the order of  $10^3$ , while the limitation in the motion range (UR3) was caused by another factor. The factor that caused said limitations is the structure of the butterfly flexure pivot itself. Unfortunately, it is highly likely that this is caused by the arbitrary values taken by the author, however no information on the butterfly model dimensions were found in multiple literature reviews. In addition, the difference in centre shift values was reviewed and no unit conversion error or other errors of any sort were found. What is very interesting here is even with the considerable error of  $10^3$ , the butterfly pivot shows very minimal shift in the CoR upon deflection. The graphs of  $D_{x_c}$  and  $D_{y_c}$  against UR3 plotted in figure (11) show that the values of centre shifts in either direction are minuscule even for precise applications. Finally, the butterfly flexure pivot structure allows it to have an almost identical centre shift no matter the region of application of the CF, this allows the pivot to be precise in multiple scenarios where the loads are not necessarily aligned with the structure perfectly.

When it comes to the compliance and stiffness of the butterfly flexure pivot, it is highly unusual that both pivots produced same magnitudes of eigenwrenches and eigentwists. Simulations were repeated multiple times but the results remain unchanged. Moreover, for LCS 1 and 2, figures (13a) and (13b) the least compliant eigenwrench ( $W_2$ ) is the one in the direction where the flexure pivot needs to be most compliant in for the rotational motion desired, while the eigentwist with greatest magnitude (least stiff to rotate around its axis) was T3 which would allow the butterfly flexure pivot to rotate in the desired direction with least stiffness. Not to mention, that T3 remains least stiff and its orientation does not change for all six LCS simulations.

## 5 CONCLUSION AND FURTHER WORK

When it comes to centre shift exclusively, the butterfly flexure pivot outweighs the CSP in the precision advantages it offers. However, when taking compliance and stiffness into account, the decision of which flexure pivot is better becomes more complex. Considering, when the LCS is that of a small load (LCS 1 and 2) the CSP shows to be more compliant in the rotation desired, but then becomes less compliant as the load increases while the opposite happens with the butterfly (assuming structure did not restrict the motion). Therefore, the author concludes that with the current shown results CSP is more reliable, but less precise. While the butterfly flexure pivot is more precise, more research needs to be done on the dimensions of its structure, as the model has shown great potential in being used for space application. In addition, further research needs to be done with the elastic limits of both flexure pivots to fully understand the range of motion and its limits. Taking into account the elastic limits of the material along with structural limits will give a clearer and more decisive conclusion on the feasibility of the pivots. Finally, with the proposed space application of solar panel pointing mechanism, the author believes the CSP would be feasible as the objective of the mechanism does not require immense precision as much as it requires a good range of motion and reliability, something the butterfly flexure pivot has not been seen to offer just yet.

## 6 ACKNOWLEDGEMENT

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